Nifty Options, India VIX and a VIX-Timed Protection Strategy

FDRM course project, Group 8, IIT Jodhpur · Oct 5, 2026

Summary

Nifty options are systematically priced above the volatility that follows, and that gap is widest right after a shock. So protection should be bought when India VIX is low and sold into spikes, never bought in panic. Using real daily Nifty 50 and India VIX data from January 2020 to April 2026, and option premiums modelled with Black-Scholes off India VIX, the main findings are below.

- **Volatility risk premium.** India VIX averaged 17.3 against 15.1 realised over the next 21 trading days, a premium of 2.2 vol points (Newey-West t = 2.8). It was positive on 78% of days and largest in the high-VIX regime (4.0 points).
- **VIX regimes.** Daily Nifty volatility was 9% annualised when VIX was below 13, 13% in the 13 to 20 band and 30% above 20. Same-day mean returns did not differ by regime (Kruskal-Wallis p = 0.66), but variances did (Levene p < 0.001). A 1% Nifty fall came with a 2.7% rise in VIX on average (correlation -0.56).
- **4 June 2024.** VIX more than doubled from 10.2 in late April to 24.6 before the result. A 30-day ATM straddle cost about 5.6% of spot against a 3.2% norm. On result day Nifty fell 5.93% (8.5% intraday), the modelled 22,500 June put rose 4.2 times, and VIX then collapsed 29% the next day.
- **Proposed hedge.** Buy a 3-month 5% out-of-the-money Nifty put on the full portfolio only when VIX is below 13. Roll it monthly while VIX stays low. Sell half the first day VIX closes at 22 or above, and the rest once the spike fades 20% from its peak.
- **Backtest.** The proposed hedge returned 13.9% a year with a -22% maximum drawdown, against 11.1% and -38% unhedged. An always-on monthly put cost 3.0% of the portfolio a year and returned 8.3%. Almost all of the advantage comes from one cheap put bought in January 2020 that was held through the COVID crash. From July 2020 onwards the timed hedge cost about 0.5% a year and did not reduce drawdowns, so it is crash insurance and not a source of extra return.

Data and method

All prices are real daily closes. Option premiums are modelled, because historical NSE option-chain data was not available for this study.

Item	Detail
Nifty 50	^NSEI daily OHLC, 17 Sep 2007 to 13 Apr 2026 (Yahoo Finance, via a GitHub mirror)
India VIX	^INDIAVIX daily OHLC, 2 Jan 2020 to 24 Aug 2026 (Yahoo Finance, via a GitHub mirror)
Common sample	1,536 trading days, 2 Jan 2020 to 13 Apr 2026
Spot check	4 Jun 2024 close: Nifty 21,884.50 and India VIX 26.75, both matching NSE
VIX regimes	Low below 13 (27% of days), Normal 13 to 20 (48%), High above 20 (25%). The cut-offs sit near the 30th and 80th percentiles.
Realised volatility	Standard deviation of daily log returns over 21 trading days, annualised with the square root of 252
Option pricing	Black-Scholes-Merton with r = 6.5% and dividend yield 1.3%. Implied vol = India VIX + 0.6 vol points per 1% out of the money (put skew). Bought at +0.5 and sold at -0.5 vol points (bid/ask).
Portfolio	100% Nifty 50 price index, starting at 1.0. Premiums are paid by selling index units, and option proceeds are reinvested the same day.

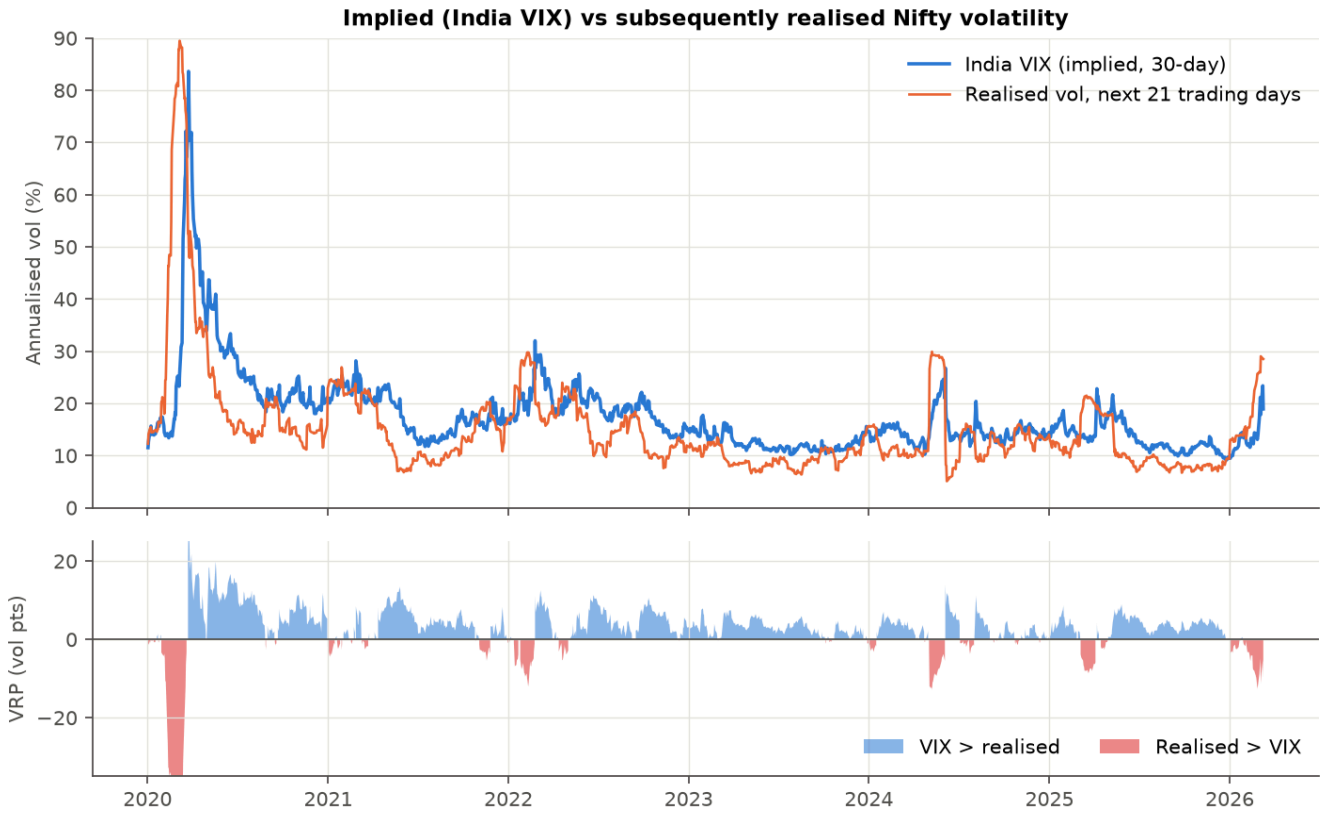
Because premiums come from India VIX, results describe how Nifty options would have been priced at VIX-consistent volatility. They do not replay actual traded prices. VIX is a 30-day measure, so using it for 3-month options ignores the term structure. That assumption and the skew are stress-tested in the robustness section.

The code, data and every table are in the GitHub repository github.com/Subham1709/FDRM_VIX. Run `fdrm_vix_analysis.py`, or open `fdrm_vix_analysis.ipynb`, to reproduce each number.

Implied vs realised volatility: the volatility risk premium

India VIX overstated the volatility that followed on 78% of days, by 2.2 vol points on average. Option sellers therefore earn a premium for bearing crash risk, and option buyers pay it. The volatility risk premium (VRP) is measured as

$$VRP_t = \text{India VIX}_t - RV_{t \to t+21}, \qquad RV = \sqrt{252}\,\operatorname{sd}(r_{t+1}, \dots, r_{t+21})$$



VIX regime	Mean VIX	Realised vol, next 21 days	Mean VRP (vol pts)	Days VRP > 0
Low (below 13)	11.7	9.5	2.1	85%
Normal (13 to 20)	15.8	14.4	1.4	75%
High (above 20)	26.5	22.5	4.0	79%
All days	17.3	15.1	2.2 (t = 2.8)	78%

Three points matter for hedging.

- **The premium is largest after shocks.** In the high regime VIX exceeded what followed by 4.0 points, because fear peaks just as volatility starts to mean-revert. That is the time to sell protection, not to buy it.
- **VIX tracks the recent past more than the future.** Its correlation with trailing 21-day realised vol is 0.86, against 0.61 with the next 21 days. A forecast regression gives RV = 1.05 + 0.81 × VIX (R^2 = 0.38), so VIX is informative but biased upward.

- **The premium flips in crashes.** The red areas in the lower panel (early 2020, early 2022, May 2024, March 2025 and early 2026) are when realised vol exceeded what was priced. Puts bought before those episodes paid off; puts bought during them were priced for it already.

India VIX: construction and use as a sentiment indicator

India VIX is NSE's estimate of expected Nifty volatility over the next 30 calendar days, in annualised percent. It is computed from the prices of out-of-the-money Nifty options rather than from any pricing model. NSE adopted the CBOE method in 2008 and computes it in real time from the order book.

1. Take near-month and next-month Nifty options. Use the best bid/ask mid-quotes of out-of-the-money puts below the forward and calls above it.
2. For each expiry, compute the model-free variance:

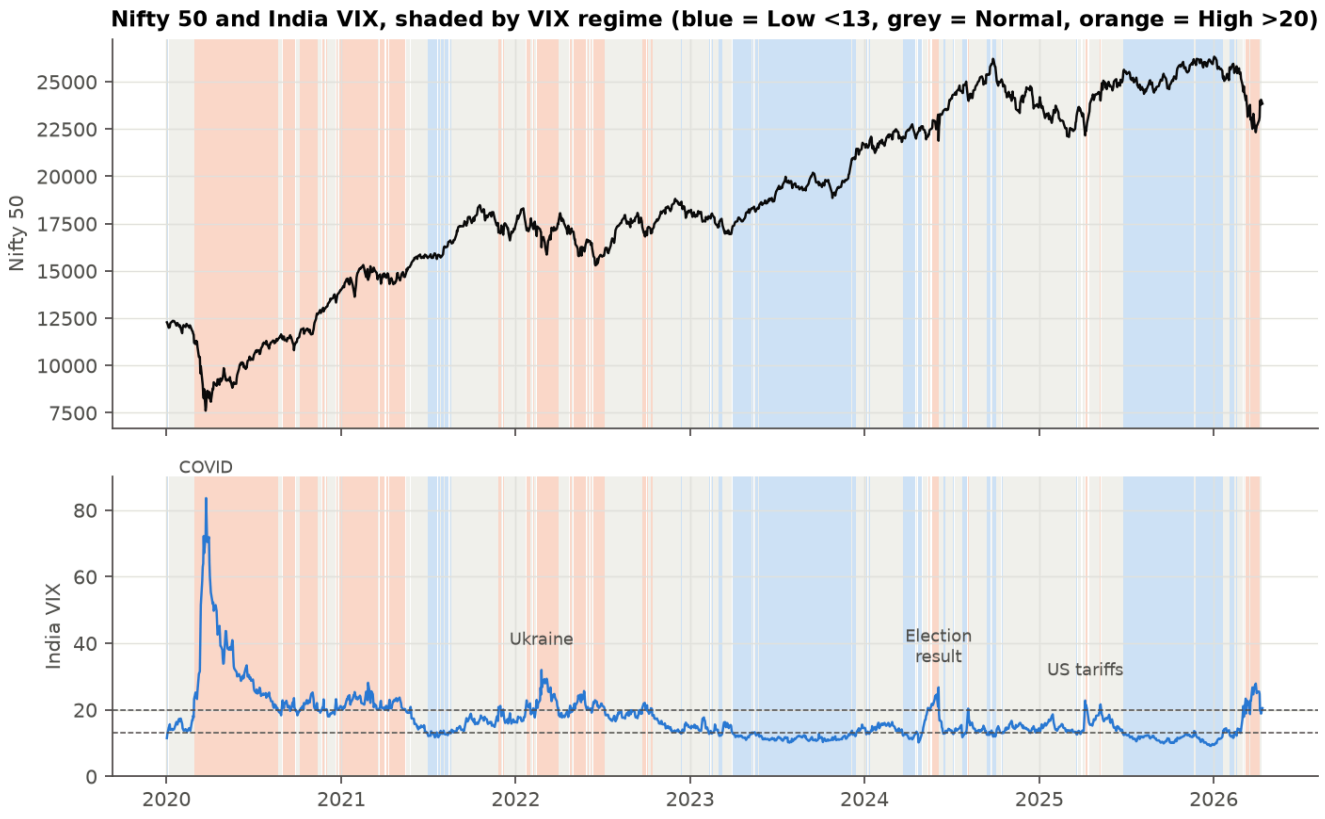
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\sigma^2 = \frac{2}{T}\sum_i \frac{\Delta K_i}{K_i^2}\, e^{RT}\, Q(K_i) \;-\; \frac{1}{T}\left(\frac{F}{K_0}-1\right)^2
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3. Here F is the forward index level implied by put-call parity, K0 is the first strike at or below F, and Q(K) is the option mid-price. T is time to expiry in minutes, expressed in years.
4. Interpolate the two variances to a constant 30-day horizon, then take India VIX = 100 × the square root of that 30-day variance.

The script reproduces this formula on a synthetic Black-Scholes chain. With a flat 15% smile it returns exactly 15.0. With put skew around a 15% at-the-money vol it returns 16.1, because the formula weights low strikes by $1/K^2$. India VIX is therefore a price of crash insurance as well as a volatility forecast.

As a sentiment gauge it works through three properties visible in the data.

- **Asymmetry (the leverage effect).** VIX rose 2.3% on the average down day and fell 1.9% on the average up day. A 1% drop in Nifty came with about a 2.7% rise in VIX.
- **Persistence, then mean reversion.** VIX regimes last weeks to months (figure below), but every spike in the sample decayed back toward 12 to 16.
- **It prices scheduled uncertainty in advance.** Before the 2024 election, VIX climbed for six weeks while Nifty barely moved.



VIX regimes and Nifty returns

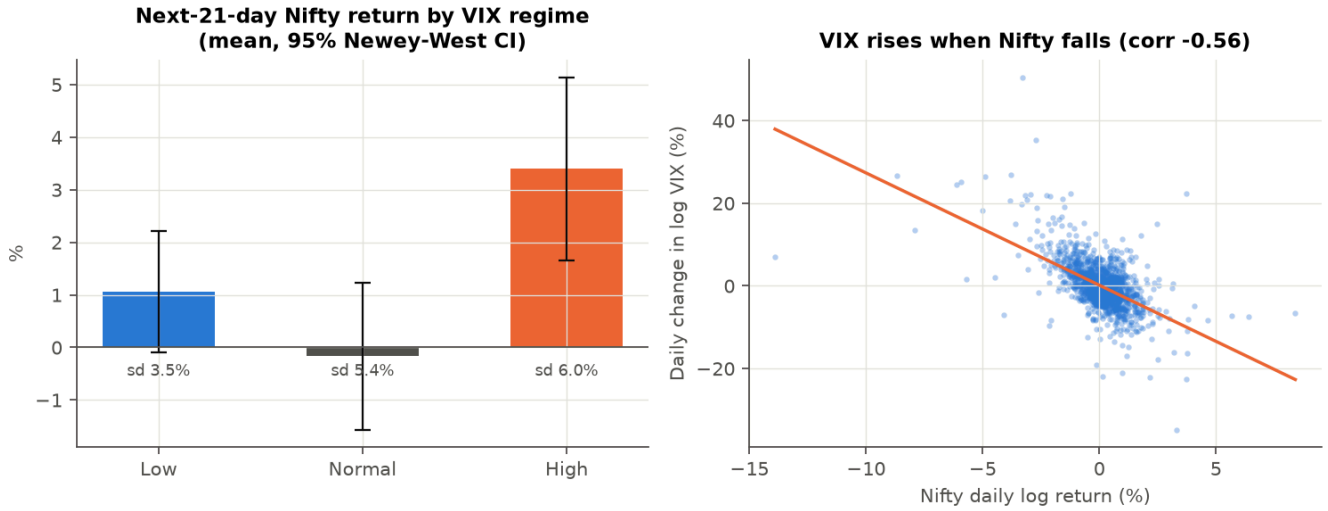
VIX regimes separate risk far more clearly than they separate return. Annualised Nifty volatility triples from the low to the high regime. Mean daily returns are statistically indistinguishable, and forward returns were highest after high-VIX days.

VIX regime	Days	Same-day vol (ann.)	Worst day	Next 21 days: mean	t (NW)	Hit rate	5th pct	Next 63 days: mean
Low (below 13)	408	9.0%	-1.4%	+1.1%	1.8	65%	-4.3%	+3.9%
Normal (13 to 20)	737	13.2%	-2.2%	-0.2%	-0.2	54%	-7.5%	+0.9%
High (above 20)	391	30.1%	-13.9%	+3.4%	3.8	76%	-4.8%	+9.5%

Forward returns overlap, so t-statistics use Newey-West standard errors with lags equal to the horizon.

- **Volatility differs, mean daily return does not.** Levene's test rejects equal variance across regimes ($p < 0.001$). Kruskal-Wallis cannot reject equal same-day return distributions ($p = 0.66$).

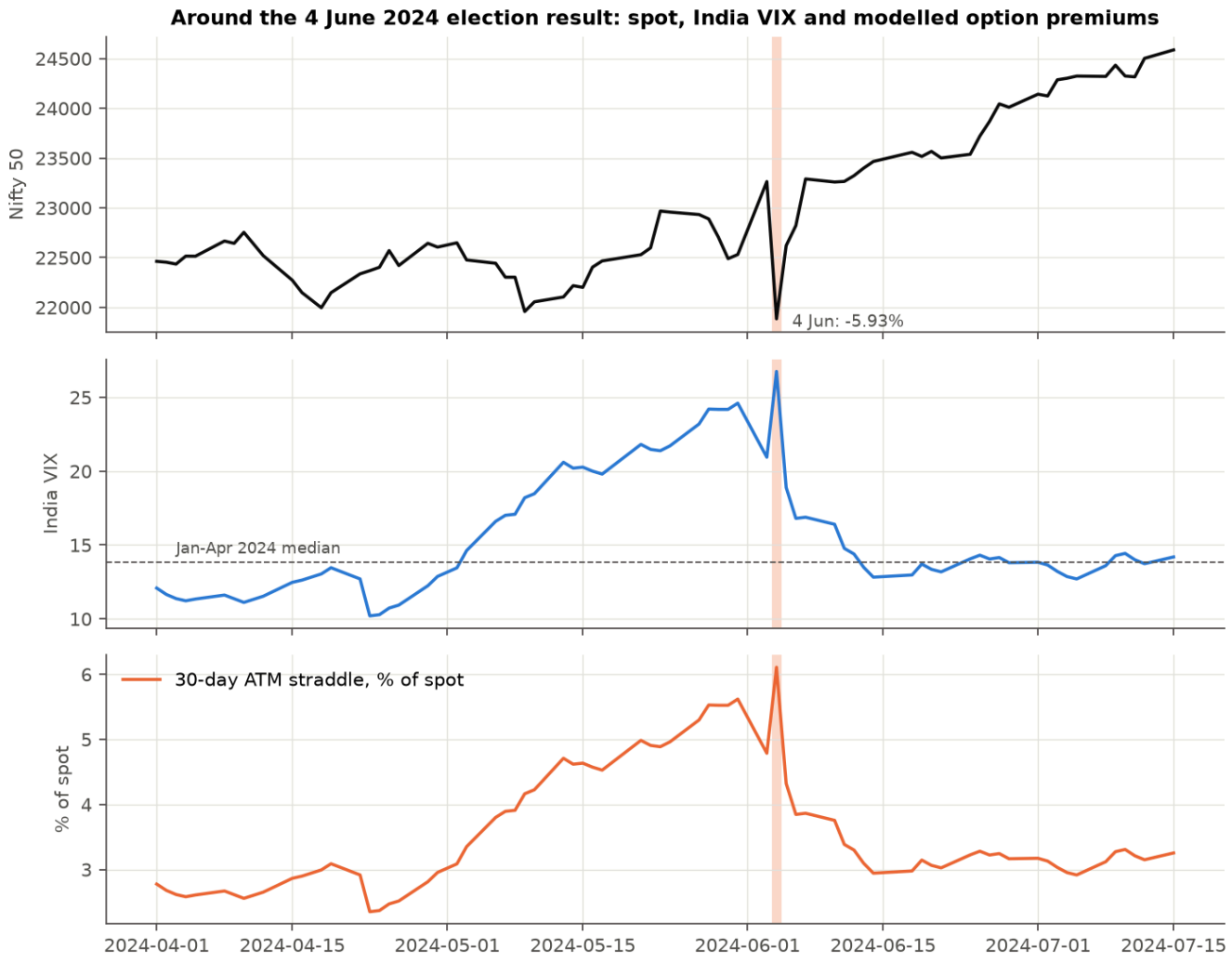
- **High VIX was followed by strong rebounds.** Regressing the next 21-day return on VIX gives +0.18% per VIX point (t = 4.3). This is mostly the 2020 recovery and is consistent with the large VRP in that regime: fear was overpriced.
- **The worst forward outcomes started in the normal regime.** Its 5th-percentile 63-day return was -14.6%. Damage tends to begin when VIX is rising from low levels, not once it is already high, so protection needs to be in place before the regime shifts.



The sample covers six years and three or four genuine stress episodes, so the regime means are imprecise. The volatility differences are robust. The return differences should be read as "no evidence that low VIX predicts losses", not as a timing signal.

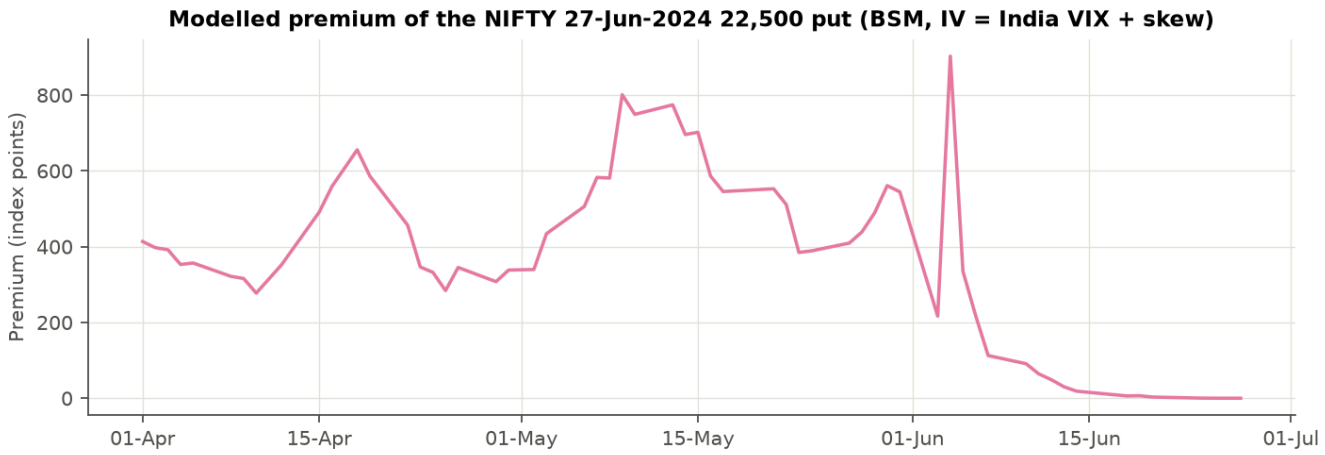
Option premiums around the 4 June 2024 election result

Around a scheduled event, premiums rise for weeks beforehand, jump again on the surprise and then collapse within a day. On a normal day they barely move. The 2024 Lok Sabha result followed this pattern exactly.



1. **Run-up (late April to 31 May).** VIX rose from 10.2 to 24.6 while Nifty traded flat between 22,000 and 23,000. A 30-day ATM straddle went from about 2.4% to 5.6% of spot, against a January to April median of 3.2%. The premium bought protection against the event, not against any change in trend.
2. **Exit polls (3 June).** Polls predicted a large majority. Nifty gained 3.25% and VIX fell 15% to 20.9, so premiums were cheapest just before the shock.
3. **Result day (4 June).** The ruling party lost its single-party majority. Nifty fell 5.93% at the close and 8.5% intraday. VIX closed 28% higher at 26.75 after touching 31.7, and the straddle rose to 6.1% of spot.
4. **Vol crush (5 to 7 June).** Nifty recovered 3.4% on 5 June and VIX fell 29% to 18.9. By 7 June VIX was 16.9 and the straddle 3.9%, close to normal.

A single contract shows the leverage. The modelled NIFTY 27 June 2024 22,500 put cost 409 points on 27 May and 217 after the exit-poll rally. It was worth 903 on 4 June, 4.2 times its value the day before, and 112 by 7 June.



The pre-event VIX told us how big the market expected the result-day move to be. If VIX in normal times was 14.4 (the January to March 2024 median) and all of the extra variance belongs to one event day, then:

$$\sigma_{event}^2 = \frac{VIX_{pre}^2 \times 30 - VIX_{base}^2 \times 29}{365}$$

This gives an expected event move of 5.8% using the 31 May VIX and 4.4% using the 3 June VIX. The actual move was 5.9%. The options market priced the election well as a single event. The 3 June exit-poll rally cheapened protection just before it was needed.

Comparison with other high-uncertainty days. The 30-day straddle is modelled from India VIX. "Move vs implied" is the absolute Nifty move divided by the previous day's VIX-implied daily standard deviation (VIX ÷ √252).

Day	Nifty move	VIX before	VIX after	VIX change	Straddle before (% spot)	Straddle after (% spot)	Move vs implied
Election result, 4 Jun 2024	-5.93%	20.9	26.8	+28%	4.8	6.1	4.6×
Exit polls, 3 Jun 2024	+3.25%	24.6	20.9	-15%	5.6	4.8	2.1×
US tariff crash, 7 Apr 2025	-3.24%	13.8	22.8	+66%	3.2	5.2	3.8×
Yen-carry unwind, 5 Aug 2024	-2.68%	14.3	20.4	+42%	3.3	4.7	3.0×
Ukraine invasion, 24 Feb 2022	-4.78%	24.5	32.0	+30%	5.6	7.3	3.2×
COVID crash, 23 Mar 2020	-12.98%	67.1	72.0	+7%	15.3	16.4	3.3×

Day	Nifty move	VIX before	VIX after	VIX change	Straddle before (% spot)	Straddle after (% spot)	Move vs implied
Normal day (median)	0.47% (absolute)	14.7	14.7	±2%	3.4	3.4	0.5×

The election stands out in two ways. VIX was already elevated before it, unlike the unscheduled shocks of 2024 and 2025, which hit when VIX was near 14. It also produced the largest move relative to implied volatility. Across the 20 largest one-day VIX jumps in the sample, VIX rose 28% on average, Nifty fell 3.2% and the 30-day straddle rose 1.3% of spot. The average absolute move on other days was 0.7%.

The proposed protection strategy

The proposal is to buy cheap, longer-dated puts only in low-VIX regimes and sell them into the spike. That turns the VRP finding into a rule: pay for convexity when it is cheapest, and hand it back when it is most overpriced.

Rule	Setting	Why
Instrument	Nifty 3-month put, strike 5% below spot, notional = 100% of portfolio	Longer tenor has lower time decay per month and more vega to gain when VIX jumps. It also survives a gap between regime signal and crash.
Entry	On the first trading day of each month, only if India VIX < 13	Low regime: options are cheapest in absolute terms and realised vol is lowest (9%)
Roll	While VIX stays below 13, sell the put each month and buy a fresh 3-month one	Keeps tenor near 3 months and strike near 5% out of the money as the market rises
No new buying	When VIX is 13 or above at the roll date	Hold the existing put to expiry, never buy in normal or high regimes, where the VRP is paid away
Monetise, first half	Sell 50% the first day VIX closes at 22 or above	Locks in the vega gain. Pre-announced events often peak before the news, as in May 2024.

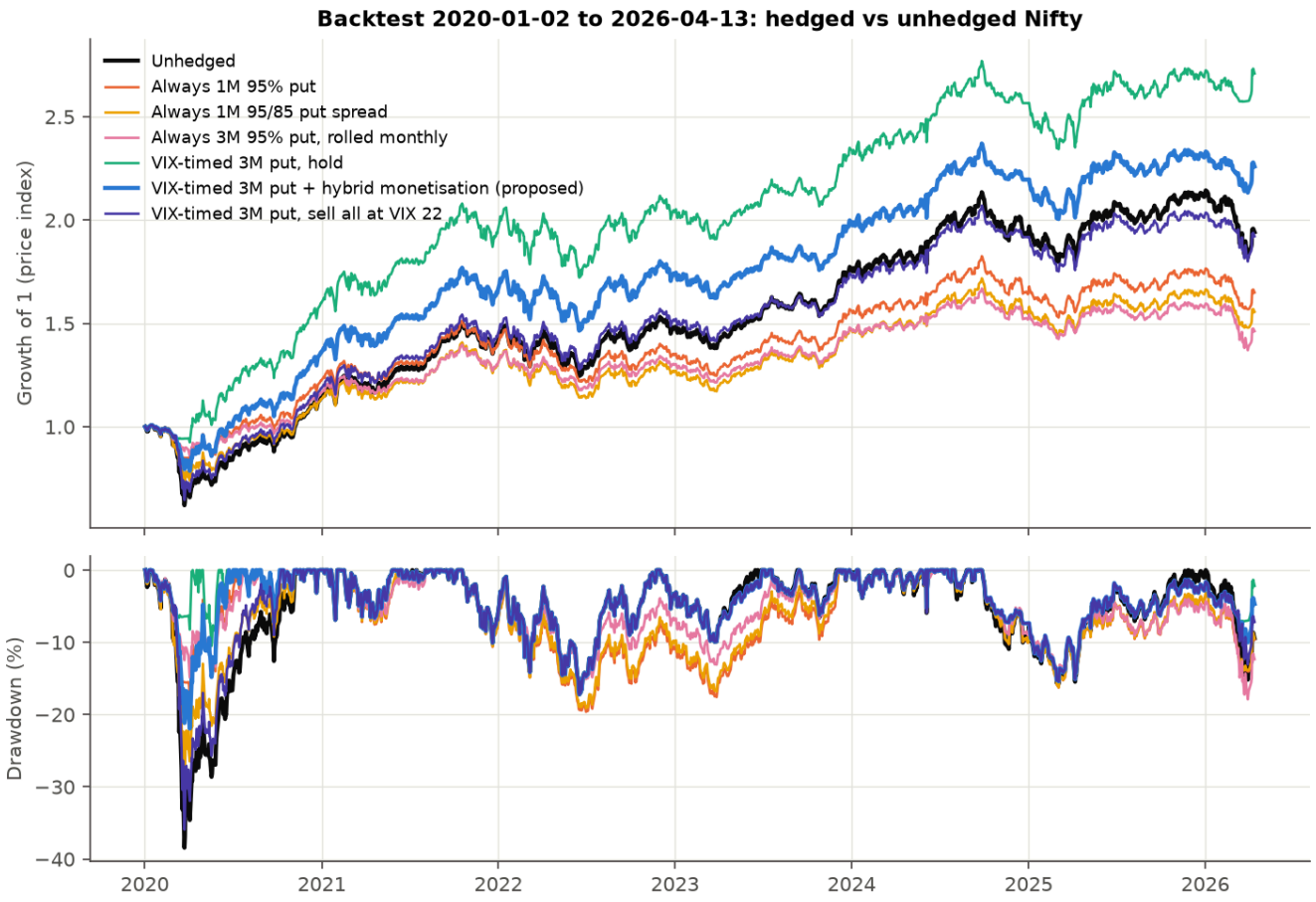
Rule	Setting	Why
Monetise, second half	Sell the rest once VIX closes 20% below its highest close since crossing 22	Lets the remaining half ride a crash that keeps going, as in March 2020. Sells it once fear starts to fade.
Use of proceeds	Reinvest in Nifty the same day	Insurance pays out when equities are cheap, which harvests the post-spike rebound seen in the regime analysis

In practice this means NSE-listed Nifty monthly and quarterly options, which are liquid at these strikes. A portfolio manager would size it as a share of equity exposure. The rule is mechanical, so it can be run from a daily VIX feed.

Two trades show how it behaves in the backtest. The put bought on 2 January 2020 at VIX 11.5 cost 93 points (0.8% of the index). Half was sold on 28 February at VIX 23 for 546 points, and the rest on 31 March at VIX 64 for 3,050 points. The put bought on 1 April 2024 (strike 21,350) cost 195 points. Half was sold on 27 May at VIX 23 for 177, during the pre-election run-up, and the rest for 95 on 5 June after the vol crush. Nifty closed above that strike even on result day, so this hedge lost a little.

Backtest: cost and payoff across VIX regimes

Over January 2020 to April 2026 the VIX-timed hedge beat both the unhedged portfolio and every always-on hedge. It held puts 45% of the time and paid 2.9% of the portfolio a year in premium, against 7.5% for a monthly put. Its edge rests mainly on the 2020 crash.

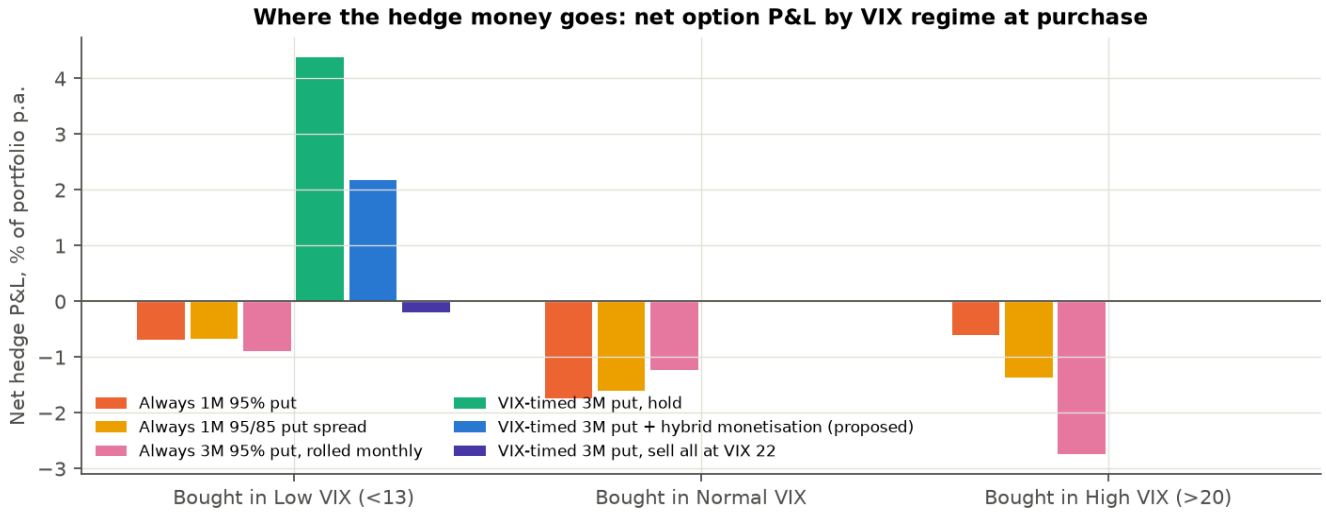


Strategy	CAGR	Vol	Sharpe	Max DD	Worst 21 days	Premium paid p.a.	Net hedge cost p.a.	Time hedged	CAGR from Jul 2020
Unhedged	11.1%	18.3%	0.25	-38.4%	-37.2%	0	0	0%	15.4%
Always 1M 95% put	8.3%	13.2%	0.14	-19.6%	-13.6%	7.5%	3.0%	99%	9.6%
Always 1M 95/85 put spread	7.3%	15.0%	0.05	-30.2%	-28.5%	9.5%	3.7%	99%	10.1%
Always 3M 95% put, rolled monthly	6.3%	11.7%	-0.02	-17.9%	-12.4%	20.2%	4.9%	97%	7.6%
VIX-timed 3M put, hold to expiry	17.2%	14.6%	0.74	-17.2%	-11.1%	2.9%	-4.4%	47%	15.2%
VIX-timed 3M put + hybrid monetisation (proposed)	**13.9%**	**15.4%**	**0.48**	**-22.0%**	**-19.5%**	**2.9%**	**-2.2%**	**45%**	**14.9%**
VIX-timed 3M put, sell all at VIX 22	11.0%	17.7%	0.25	-35.9%	-34.6%	2.9%	0.2%	43%	14.4%

Sharpe uses a 6.5% risk-free rate. Premium and net cost are option cash flows as a percentage of portfolio value, averaged per year. A negative net cost means the options returned more than they cost. All returns exclude dividends.

Cost and payoff by VIX regime. The next table attributes each option's whole life (premium paid less money received) to the VIX regime on the day it was bought, in % of the portfolio per year. The chart below shows the same numbers.

Strategy	Bought in Low VIX	Bought in Normal VIX	Bought in High VIX
Always 1M 95% put	-0.7%	-1.7%	-0.6%
Always 1M 95/85 put spread	-0.7%	-1.6%	-1.4%
Always 3M 95% put, rolled monthly	-0.9%	-1.2%	-2.7%
Proposed (only buys in Low)	+2.2%	none bought	none bought



Viewed day by day, the proposed hedge cost 2.1% a year relative to unhedged on low-VIX days, the insurance carry. It earned 2.0% a year on normal days and 6.3% on high-VIX days. The always-on monthly put lost 9.2% a year on high-VIX days, because it kept buying expensive protection just before rebounds.

Stress episodes (portfolio return over each window):

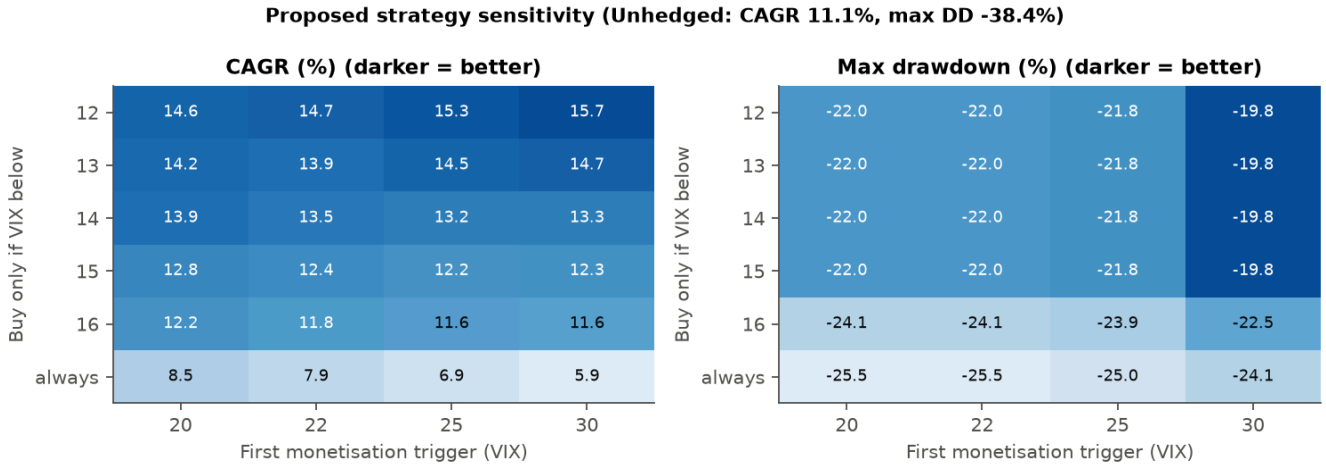
Episode	Unhedged	Always 1M put	Always 3M put	Proposed	Timed, hold	Timed, sell at 22
COVID crash, 17 Jan to 23 Mar 2020	-38.4%	-15.5%	-12.4%	-21.1%	-6.3%	-35.9%

Episode	Unhedged	Always 1M put	Always 3M put	Proposed	Timed, hold	Timed, sell at 22
War and rate hikes, 17 Jan to 17 Jun 2022	-16.5%	-17.1%	-13.9%	-16.5%	-16.5%	-16.5%
Election result, 3 to 4 Jun 2024	-5.9%	-3.6%	-3.1%	-5.2%	-4.5%	-5.9%
Yen-carry unwind, 1 to 5 Aug 2024	-3.8%	-2.6%	-1.9%	-1.9%	-1.9%	-1.9%
Slow correction, 26 Sep 2024 to 4 Mar 2025	-15.8%	-16.3%	-14.7%	-15.4%	-15.4%	-15.4%
Tariff shock, 28 Mar to 7 Apr 2025	-5.8%	-4.3%	-3.4%	-5.8%	-5.8%	-5.8%

The timed hedge protects only when a shock arrives from a calm, low-VIX market, as in January 2020 and August 2024. It gives no protection in a slow grind (2024 to 2025) or when VIX is already in the normal band before the drop (2022, April 2025). Always-on puts protect more often, but they cost 3 to 5% a year, which is more than they saved over this period.

Robustness, limitations and conclusion

Buying only in low VIX holds up across thresholds, skew and spread assumptions. Two other conclusions are fragile: the size of the gain and the exact monetisation rule. Both hinge on whether a cheap put happened to be alive when COVID hit.



- **Entry threshold.** Buying only below a VIX of 12 to 15 gave a CAGR of 12.2% to 15.7% and maximum drawdowns of -20% to -22%. Buying regardless of VIX gave 5.9% to 8.5%. This is the most robust result.
- **Pricing assumptions.** Across skew of 0 to 1.0 vol point per 1% OTM and half-spreads of 0 to 1.0 point, the proposal returned 12.9% to 15.1%. The always-on monthly put returned 6.0% to 11.1%. Drawdown stayed between -21.7% and -22.2%.

- **Tenor is fragile.** 1-month and 2-month versions returned about 10.5% with -36% to -39% drawdowns. A 3-month or 6-month put returned about 13.7% with a -22% drawdown. The January 2020 2-month put expired on 2 March, a week before the crash accelerated. That is luck of the calendar, and it is the strongest argument for longer tenors.
- **The monetisation rule is fragile.** With 3-month puts, holding to expiry returned 17.2%, the trailing rule 16.5%, the hybrid 13.9%, the 22/30 ladder 11.9% and selling everything at 22 just 11.0%. Selling early wins in short, scheduled spikes, and holding wins in a crash. The hybrid is chosen because it does reasonably in both and no one knows in advance which kind of spike it is.
- **Strike.** At-the-money puts returned 15.3% (-18.6% drawdown). 10% OTM puts returned 13.0% (-24.9%).
- **Without the COVID crash** (July 2020 to April 2026), no hedge beat unhedged. The proposal returned 14.9% against 15.4%, with the same -17.2% maximum drawdown. Always-on puts returned 7.6% to 10.1%. Over a calm period the timed hedge costs about 0.5% a year, while always-on protection costs 5 to 8%.

Limitations.

1. Premiums are modelled from India VIX, not traded prices. In calm markets the real Nifty term structure slopes upward and OTM put skew is steeper, so real 3-month puts bought at low VIX probably cost more than modelled. The skew and spread sensitivities bound that, but they do not remove it.
2. India VIX history available here starts in 2020. The sample contains one true crash, and the regime thresholds were set on the same data they are tested on.
3. Option expiries are synthetic (the first trading day of each month), not NSE's actual expiry days. The results show how much expiry timing can matter.
4. Prices exclude dividends, taxes, STT, brokerage, margin and slippage beyond the 0.5 vol point spread. Execution is at the daily close.

Conclusion. Nifty option buyers pay a volatility risk premium of about 2 vol points, rising to about 4 points just after a shock. India VIX is a reliable real-time gauge of that price and of market fear. It is not a reliable signal of direction. The 4 June 2024 result shows the typical event pattern: premiums build for weeks, spike on the surprise and collapse within a day. Protection therefore pays best when bought before the build-up and sold during it. A long-term Nifty portfolio can carry tail protection for about 0.5% a year by buying 3-month 5% OTM puts only when VIX is below 13 and monetising them on a spike. Over 2020 to 2026 that cut the maximum drawdown from 38% to 22%. Investors should treat it as cheap crash insurance that pays rarely, not as a source of return.